Beyond age and sex:

Using multi-stakeholder perspectives to define meaningful subgroups for fairness assessment in clinical prediction modelling

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1. Background

- **Clinical prediction models** (CPMs) use **patient characteristics** (e.g., age, sex, ethnicity) to **estimate the risk of future outcomes** [1].
- There is **increasing concern** that CPMs perform **unfairly for some patients**, and **limited guidance** on defining **clinically meaningful subgroups beyond traditional categories** or **accounting for intersectionality** [2].
- **RENAL-HF** is a **personalised renal function monitoring tool** developed to **support patient monitoring** and **aid clinical decision-making** in **patients with heart failure** (HF) [3, 4].
- The **foundation of the tool** is a **longitudinal CPM** which **predicts a patient's kidney function** based on several predictors [5].

How can we identify potential subgroups to assess performance and fairness of CPMs?

2. Methods

- We **conducted** a **scoping review** to **identify** the **most appropriate fairness definitions**, **concerns**, and **assessment methods** to **inform the fairness evaluation of RENAL-HF**.
- We explored **participatory, stakeholder-informed approaches** to **subgroup identification and fairness definitions**, using **multi-stakeholder perspectives** from patients, carers, and healthcare professionals to **develop patient personas** for RENAL-HF [6].

3. Results: Scoping review

Review focused on CPMs for heart failure outcomes

1,186 articles screened → 14 included in review



Fairness definitions were limited and inconsistent across studies



Assessment approaches varied, with subgroup performance most common



Race, sex, and age most commonly assessed characteristics

4. Results: Patient personas

Secondary analysis of qualitative data collected during model development

Patient personas were developed to showcase examples of HF patients and associated scenarios that need to be considered to avoid unintended and inequitable consequences

Stakeholder surveys and a workshop were carried out with patients, carers and healthcare professionals (e.g. nurses, pharmacists, GPs) to refine the personas and scenarios

9 patient personas developed:

Capturing variation in symptoms, lifestyle, disease progression, and condition knowledge

Underrepresented subgroups identified:

Including younger HF patients, patients from deprived areas, and patients with substance use issues

Potential fairness concerns discussed:

Related to differences in healthcare engagement and societal factors

5. Conclusion

- There is **growing interest** in **fairness in CPMs** and in **patient and stakeholder input into CPM research**. However, there is **limited guidance** on **how to use these perspectives in practice**.
- This study **provides a potential approach** to using **multi-stakeholder perspectives in CPM research** to **define meaningful subgroups** for **fairness assessments**.
- **Beyond race, sex, and age**, **additional sociodemographic factors** and **intersections between characteristics** may **be important** for **fairness evaluation** in HF and other contexts.
- Although some groups may not be well represented in available data, this approach provides a **reproducible framework** to support **broader inclusion of vulnerable or marginalised groups** and **enable** more **robust and sustainable fairness assessment of CPMs**.

6. Next steps

- Explore the **fairness** of the **RENAL-HF tool**.
- Investigate **sample size requirements** and **missing data approaches** to ensure **accurate and fair longitudinal CPMs**.

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